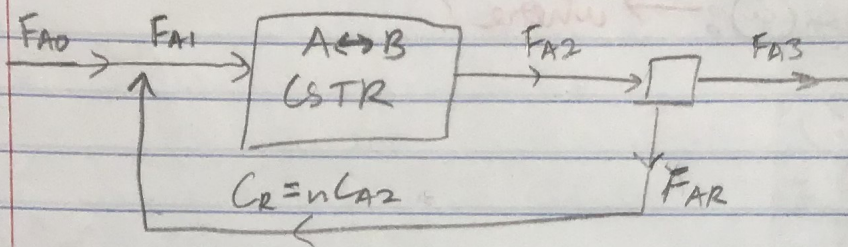


CHE 321 - HW 2

(a) $A \rightleftharpoons B$ liquid 1st order $K=1$, so $k_1 = k_{-1}$ 

$$F_{A0} + F_{AR} = F_{A1}$$

$$F_{A0} = F_{A1} - F_{AR}$$

$$F_{A2} = F_{AR} + F_{A3}$$

$$F_{AR} = F_{A2} - F_{A3}$$

$$F_{A2} = F_{A1}(1-x)$$

$$x_T = 1 - \frac{F_{A3}}{F_{A0}}$$

$$F_{A0} = F_{A1} - F_{A2} + F_{A3}$$

$$F_{A3} = F_{A2}(1-\alpha)$$

$$F_{A1} = F_{A0} - F_{AR} = F_{A0} + \alpha F_{A2}$$

where $\alpha = \frac{F_{AR}}{F_{A2}}$ or $\alpha = \frac{C_{A2} V_R}{C_{A2} V_2 + V_0 + V_R}$

$$F_{A2} = F_{A1}(1-x)$$

$$F_{A0} = F_{A1}(1-\alpha+x)$$

$$F_{A2} = F_{A0}(1-x)$$

$$r_A = -k_1 C_A + k_{-1} C_B$$

$$\text{in} - \text{out} + \text{gen} = 0$$

$$F_{A1} - F_{A2} + (k_1 C_{A2} + k_{-1} C_{B2}) V = 0$$

$$F_{A1} x + (k_1 \frac{F_{A2} x}{V_2} + k_{-1} \frac{F_{A1} x}{V_1}) V = 0$$

$$F_{A1} x + k_1 \left(\frac{F_{A2} + F_{A1} x}{V_2} \right) V = 0$$

$$C_{B2} = \frac{C_{A1} x}{V_1} = \frac{F_{A1} x}{V_1}$$

$$\frac{F_{A0} x}{1-\alpha+x} + k_1 \left(\frac{F_{A1}(1-x)}{V_2} + \frac{F_{A1} x}{V_1} \right) V = 0$$

$$\frac{F_{A0} x}{1-\alpha+x} + k_1 \left(\frac{F_{A0} x}{(1-\alpha+x)V_1} - \frac{F_{A0} F_{A0} x}{(1-\alpha+x)V_1} \right) V = 0$$

$$\frac{F_{A0} x}{1-\alpha+x} + \frac{k_1 V}{V_1} \left(\frac{2F_{A0} x - F_{A0}}{(1-\alpha+x)} \right) = 0$$

$$x + \frac{2k_1 V}{V_1} - \frac{k_1 V}{V_1} = 0$$

$$x \left(1 + \frac{2k_1 V}{V_1} \right) = \frac{k_1 V}{V_1}$$

$$x = \frac{k_1 V}{V_1}$$

$$x = \frac{k_1 V}{V_1 + 2k_1 V}$$

$1 + \frac{2k_1 V}{V_1}$ combine

(b) Without recycle stream:

$$V_R = 0$$

Using eqn from (a) $\rightarrow$ where?

$$x = \frac{L_1 V}{V_1 + 2L_1 V} \quad (?)$$